9

Energy, Power and Resistance

e.m.f. and p.d.
Power
Resistance
Resistivity
Superconductors

Circuit symbols

Description	Symbol
conductors crossing with no connection	
junction of conductors	
open switch	
closed switch	
open push switch	
closed push switch	
cell	
battery of cells	
power supply	+ − (dc) or (ac)
transformer	
ammeter	A
milliammeter	mA
voltmeter	V
fixed resistor	
variable resistor	

Description	Symbol
heater	
thermistor	
light-dependent resistor (LDR)	
relay	
diode	
light-emitting diode (LED)	
lamp	
loudspeaker	
microphone	
electric bell	
earth or ground	
motor	M
generator	G
fuse/circuit breaker	

potential divider

e.m.f. and pd

Potential difference

Work done by charge carrier per unit charge between two points

It is a measure of the energy transferred by charge carriers

Unit - volts

e.m.f ($\mathcal{E}$)

Work done on charge carriers per unit charge

It is a measure of the energy transferred on to charge carriers by an energy source

Unit - volts

Voltmeters theoretically should have infinite resistance so minimal current flows through voltmeter

Electron guns in cathode ray tubes:

- Electrons can be emitted from a metal surface if it is hot enough (thermionic emission)
- In a vacuum, electrons can be accelerated across a potential difference created by a cathode and anode

$$eV = \frac{1}{2}mv^2$$

Power

$$P = \frac{W}{t} \qquad P = I^2R \qquad P = \frac{V^2}{R}$$

1 kWh → energy transferred by a 1 kW device over 1 hour

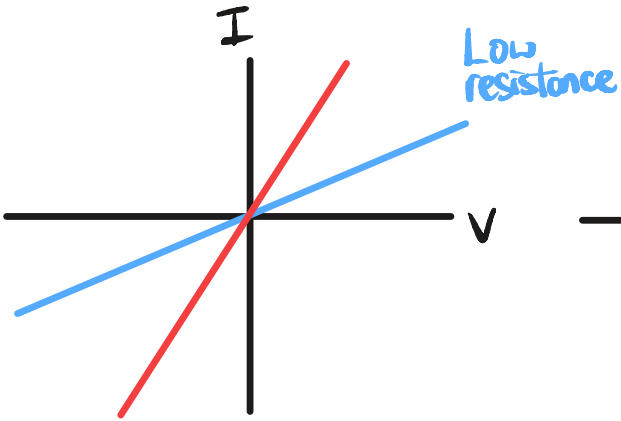
Resistance

Definition: the potential difference required for one unit of current to flow through the component
Unit - ohm

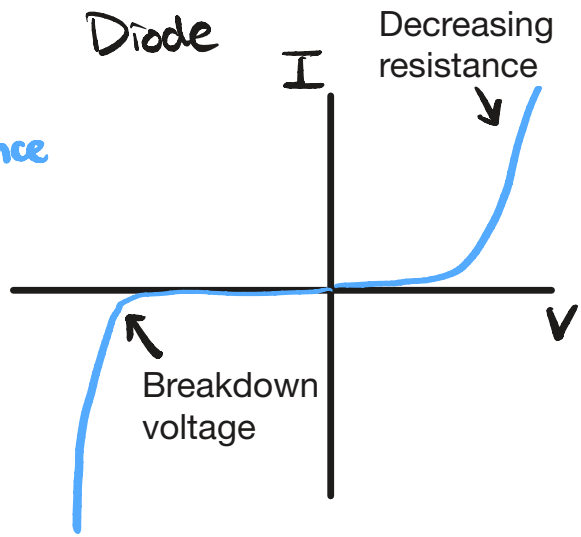
Ohm's law: the current through a conductor between two points is directly proportional to the potential difference across the two points provided that the temperature remains constant

$$\Delta T = 0, \quad V \propto I$$

Fixed resistor



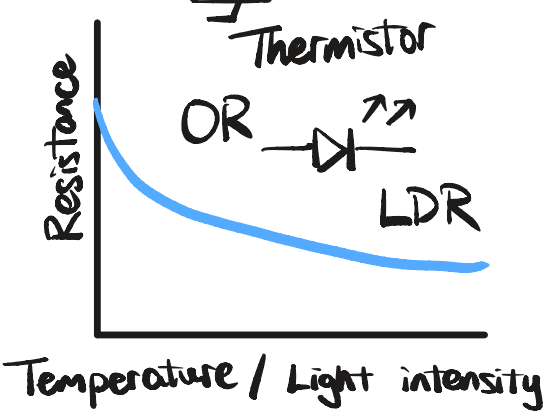
Diode



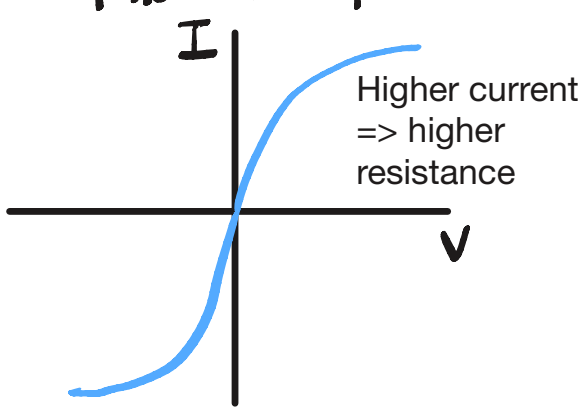
Thermistor

OR

LDR



Filament lamp



Resistivity

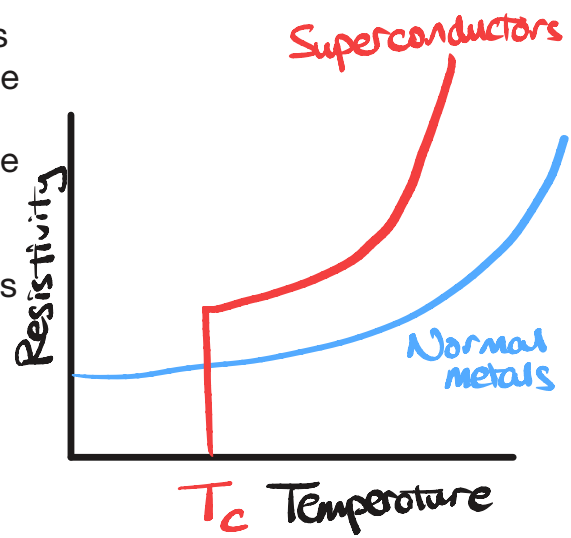
Denoted by rho (ρ)
Intrinsic property of a material that quantifies how strongly it opposes the flow of electric current
Electrical resistance (R) of a conductor with a unit cross-sectional area (A) and unit length (L)
For a material, resistivity is a constant at a given temperature
Unit - Ωm

$$\rho = \frac{RA}{L}$$

Superconductors

Temperature increases
=> Lattice atoms/ions vibrate faster and more vigorously
=> Free electrons collide with lattice atoms more often
=> Electrons' rms speed increases with temperature
=> Drift velocity decreases

In some materials, when the material is cooled to a critical temperature (Tc), the resistance of the material falls the zero
The state of zero resistance is when the material becomes superconducting



The limiting factor of a semiconductor's resistivity is the number density of charge carriers

For semiconductors:
Higher temperature
=> Higher energy
=> number density increases
=> resistivity decreases